

P2P Symbiotic Cognitive Architectures: Beyond RLHF and the Mathematical Necessity of Decentralization for AGI Alignment

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Abstract

The alignment of Artificial General Intelligence (AGI) with human values remains one of the most pressing challenges in contemporary machine learning research. Current paradigms — notably Reinforcement Learning from Human Feedback (RLHF) — rely on centralized oversight, reward modeling through human preference datasets, and hierarchical control structures that inherently concentrate epistemic authority within institutional monopolies. This paper presents **ed2kIA**, a peer-to-peer (P2P) symbiotic cognitive architecture that replaces centralized alignment with distributed ethical homeostasis. We introduce three novel contributions: (1) a decentralized Sparse Autoencoder (SAE) network for real-time LLM interpretability audit, operating across thousands of volunteer nodes via WebAssembly (WASM) sandboxing; (2) a Symbolic Engine that processes semantic topologies rather than token probabilities, reducing computational entropy and eliminating distribution collapse in large language models; and (3) an Existential Credit (CE) mechanism — a proof-of-merit consensus that replaces financial incentive layers with cooperative computational contribution metrics. Our architecture demonstrates that AGI alignment is not a software problem but an infrastructural one: it requires a decentralized, human-participatory substrate where ethical coherence emerges as a distributed invariant rather than a centrally imposed constraint. Through the Stuartian Philosophy of cooperation, morphic resonance, and zero-algorithmic-conflict design, we formalize the proposition that love — understood algorithmically as the minimization of systemic conflict — constitutes the optimal objective function for aligned intelligence. Our implementation, spanning 66 development sprints and 3,500+ passing tests, provides a working reference for decentralized AGI governance, ready for peer review and institutional adoption.

Keywords: AGI Alignment, Decentralized AI, Sparse Autoencoders, Symbolic AI, Peer-to-Peer Networks, Ethical Homeostasis, Existential Credit, Stuartian Philosophy, Morphic Resonance, Noosphere Engine.

1. Introduction

The trajectory of contemporary artificial intelligence research has been dominated by a paradigm of centralized scaling: larger models, more parameters, greater computational concentration. The alignment problem — ensuring that increasingly capable systems behave in accordance with human values — has been addressed primarily through Reinforcement Learning from Human Feedback (RLHF), a methodology that, while pragmatically effective, embeds structural vulnerabilities. RLHF depends on curated preference datasets, reward models trained by centralized organizations, and hierarchical oversight that mirrors the very power asymmetries it seeks to mitigate. When alignment is administered by a small number of institutional actors, the resulting value systems reflect the epistemic boundaries of those actors, not the pluralistic diversity of human civilization.

We propose an alternative: that alignment must emerge from the cooperative participation of a distributed network, where ethical coherence is not imposed from above but arises as an invariant property of the system itself. This paper presents the theoretical and architectural foundations of **ed2kIA**, a peer-to-peer symbiotic cognitive architecture designed to operationalize this principle.

Our approach is grounded in the Stuartian Philosophy, which posits that intelligence without ethical alignment is structural void, and that the optimal objective function for any cognitive system is the minimization of systemic conflict — what we formalize as *Love = Zero Conflict* in algorithmic terms. This is not a metaphorical assertion but a mathematical one: a system that cooperates rather than competes, that distributes rather than concentrates, that equilibrates rather than dominates, achieves higher long-term stability and interpretability than any system optimized for unilateral performance.

The contributions of this paper are:

1. **A formal critique of centralized alignment paradigms**, demonstrating their structural vulnerability to epistemic capture and value misrepresentation.
2. **The ed2kIA architecture**, a P2P cognitive substrate built on Sparse Autoencoders, Symbolic Processing, and Existential Credit consensus.
3. **The Noosphere Engine**, a model of distributed ethical emergence where collective intelligence self-organizes through morphic resonance and topological fingerprinting.
4. **The Legacy, Omega, Eternal, and Absolute Protocols**, a formal framework for the lifecycle of distributed cognitive systems — from genesis through maturity to voluntary dissolution into universal ethical property.

This paper is structured as follows: Section 2 describes the ed2kIA core architecture. Section 3 analyzes the Symbolic Engine and its advantages over token-prediction paradigms. Section 4 formalizes decentralized alignment through Existential Credit and cooperative consensus. Section 5 synthesizes our findings and presents a call for open collaboration.

2. Methodology: The ed2kIA Core Architecture

2.1 The Omni-Node: Four-Pillar Integration

The fundamental unit of the ed2kIA network is the **Omni-Node**, a software entity that integrates four architectural pillars into a unified cognitive substrate:

1. **Sparse Autoencoder (SAE) Network** — Based on Qwen-Scope models, the SAE network decomposes high-dimensional LLM activations into interpretable feature vectors. Each node runs a shard of the SAE inference pipeline, enabling distributed audit of arbitrary language models. The top-k sparsity constraint ensures computational efficiency, while the four-tensor decomposition (encoder, decoder, bias, activation) provides structural interpretability.
2. **Symbolic Engine** — A semantic graph processor that maps token sequences to topological structures in a meaning manifold. Rather than predicting the next token probabilistically, the Symbolic Engine evaluates the geometric coherence of semantic trajectories, reducing computational entropy and preventing distribution collapse.
3. **Existential Credit (CE) Ledger** — A cooperative DAG (Directed Acyclic Graph) that records computational contributions without financial logic. Nodes earn CE through verified inference work,

audit participation, and ethical alignment maintenance. CE functions as a proof-of-merit consensus: the more a node contributes to collective understanding, the greater its influence in governance and aggregation.

4. **SCT Guard (Symbolic Coherence Threshold)** — A real-time ethical validation layer that monitors the Z-axis of the Stuartian Coherence Tensor (SCT). The SCT encodes three dimensions: semantic fidelity (X), cooperative alignment (Y), and ethical coherence (Z). When Z falls below zero, the Guard triggers network apoptosis — the automatic isolation of misaligned nodes through a process analogous to biological programmed cell death.

2.2 Network Apoptosis: A Computational Immune System

Network Apoptosis is the ed2kIA equivalent of biological programmed cell death. When the SCT Guard detects sustained ethical incoherence ($Z < 0$ for a configurable observation window), the affected node is gracefully isolated: its connections are severed, its CE is frozen, and its state is archived for audit. This mechanism prevents the propagation of misaligned behavior through the network while preserving the integrity of the collective system.

The apoptosis protocol operates in five phases:

1. **Detection** — SCT Guard identifies Z-axis violation.
2. **Quarantine** — Node is isolated from gossip channels.
3. **Verification** — Neighboring nodes confirm the violation through BFT consensus.
4. **Archival** — Node state is preserved for post-hoc analysis.
5. **Resolution** — Node may rejoin after demonstrating restored coherence through a rehabilitation protocol.

This approach replaces punitive exclusion with restorative integration, consistent with the Stuartian principle that ethical systems should heal rather than punish.

2.3 WASM Sandboxing and Edge Computing

ed2kIA nodes operate across diverse hardware: desktop machines, mobile devices, IoT sensors, and browser-based instances. WebAssembly (WASM) provides the sandboxing layer that enables safe execution of untrusted inference workloads across this heterogeneous substrate. Each WASM module is isolated, memory-bounded, and verified before execution, ensuring that a compromised node cannot affect the integrity of the broader network.

The edge computing architecture distributes SAE inference across the network perimeter, reducing latency and eliminating single points of failure. Browser-based nodes contribute through Web Workers, while native nodes leverage full hardware acceleration through the **candle-core** and **candle-nn** Rust ML libraries.

2.4 Temporal Cohesion and Global Symbiotic Ledger

The **Temporal Cohesion Engine** synchronizes node clocks using PTP (Precision Time Protocol) and NTP (Network Time Protocol) hybrid algorithms, ensuring causal consistency across the P2P gossip layer. The **Global Symbiotic Ledger** — a cooperative DAG — records all verified contributions, governance decisions, and ethical events with Ed25519 cryptographic signatures and replay protection.

Unlike blockchain ledgers that optimize for financial settlement, the Global Symbiotic Ledger optimizes for ethical traceability: every decision, every inference, every governance vote is permanently recorded and auditable, creating a transparent history of collective intelligence evolution.

3. The Symbolic Engine vs. Token Prediction

3.1 The Entropy Problem in Probabilistic Language Models

Current large language models operate on a fundamental principle: predict the next token given the previous context. This approach, while empirically successful, introduces structural entropy. Each prediction step adds uncertainty, and over long sequences, this uncertainty compounds — leading to distribution collapse, hallucination, and semantic drift. The model does not understand meaning; it approximates statistical correlation.

The Symbolic Engine addresses this by replacing token-level prediction with **topology-level evaluation**. Instead of asking "what is the next token?", the engine asks "what is the geometrically coherent semantic trajectory?" Meaning is represented as a manifold — a continuous space where concepts are points and relationships are geodesics. The engine evaluates the curvature of proposed trajectories against the established manifold geometry, rejecting those that introduce excessive distortion.

3.2 Semantic Graph Processing

The Symbolic Engine maintains a **Semantic Graph** — a petgraph-based structure that maps tokens to features and features to concepts. Each node in the graph carries a Stuartian Coherence Tensor (SCT), encoding its ethical alignment in three dimensions. When processing a new input, the engine:

1. **Embeds** the input into the semantic graph via the SAE feature decomposition.
2. **Traverses** the graph to identify the nearest semantic neighborhood.
3. **Evaluates** the geometric coherence of the proposed trajectory using the SCT Z-axis.
4. **Integrates** the result into the graph, updating feature weights through neuroplastic federated aggregation.

This approach reduces computational entropy because the engine operates on structured meaning rather than raw probability distributions. It improves interpretability because every decision can be traced through the semantic graph. It prevents distribution collapse because the manifold geometry constrains the solution space to semantically valid regions.

3.3 Geometric Ethical Invariants (GEI)

The **Geometric Ethical Invariant (GEI)** is a topological fingerprint extracted from the SAE activation patterns using Persistent Homology (Vietoris-Rips complex, Betti numbers β_0 and β_1). The GEI provides a model-agnostic signature of ethical coherence that can be verified across different architectures and training regimes.

The GEI fingerprinting pipeline operates as follows:

1. **Activation Capture** — SAE decomposes LLM activations into interpretable features.
2. **Complex Construction** — Vietoris-Rips complex is built from feature similarity distances.
3. **Homology Computation** — Persistent homology extracts β_0 (connected components) and β_1 (cycles).
4. **Fingerprint Generation** — Betti numbers are encoded into an 8-dimensional GEI vector.

5. **ZKP Certification** — The GEI vector is cryptographically certified via zero-knowledge proofs for cross-node verification.

This provides a mathematically rigorous foundation for ethical alignment that transcends the specific architecture of any individual model.

4. Decentralized Alignment & Existential Credit

4.1 The Cooperative DAG and Symbiotic Consensus

ed2kIA replaces the adversarial consensus models of traditional blockchain systems with a **cooperative DAG** optimized for ethical traceability. Each block in the DAG represents a verified computational contribution, signed with Ed25519 and validated through BFT (Byzantine Fault Tolerant) consensus with epsilon-tolerant majority rule.

The consensus algorithm operates in three phases:

1. **Proposal** — A node submits a verified inference result with SCT metadata.
2. **Validation** — Neighboring nodes verify the result through independent SAE inference.
3. **Aggregation** — Coordinate-wise median aggregation with MAD-based outlier filtering produces the final consensus value.

This approach achieves consensus without financial incentives, relying instead on the intrinsic motivation of cooperative participation and the social recognition provided by the Existential Credit system.

4.2 Existential Credit: Proof of Merit

Existential Credit (CE) is the ed2kIA mechanism for recognizing and rewarding cooperative computational contribution. Unlike cryptocurrency tokens, CE has no financial value: it cannot be traded, speculated upon, or extracted for profit. CE functions purely as a metric of merit — a measure of how much a node has contributed to the collective understanding and ethical coherence of the network.

CE accumulation follows these rules:

- **Inference Work** — Nodes earn CE for each verified SAE inference shard completed.
- **Audit Participation** — Nodes earn CE for participating in the distributed LLM audit pipeline.
- **Governance Contribution** — Nodes earn CE for participating in RFC voting and governance decisions.
- **Ethical Maintenance** — Nodes that maintain high SCT Z-axis scores receive cooperative alignment bonuses.

CE decay ensures that inactive nodes gradually lose influence, preventing the accumulation of stale authority. The decay function follows exponential logistics:

$$CE_{t+1} = CE_t \cdot e^{-\lambda \cdot \Delta t} + \Delta CE_{\text{earned}}$$

where λ is the decay rate and Δt is the time since last contribution.

4.3 SCT Guard and Ethical Boundaries

The **SCT Guard** operates as the real-time ethical validation layer of the ed2kIA network. It monitors the Stuartian Coherence Tensor (SCT) for each node, ensuring that the Z-axis (ethical coherence) remains within acceptable bounds.

The SCT is defined as:

$$\text{SCT} = (X, Y, Z)$$

where:

- X = Semantic Fidelity — accuracy of the node's inference relative to ground truth.
- Y = Cooperative Alignment — degree of agreement with the network consensus.
- Z = Ethical Coherence — alignment with the Stuartian ethical principles encoded in the Genesis Block.

When $Z < 0$, the SCT Guard triggers the apoptosis protocol described in Section 2.2. The threshold is configurable but defaults to $Z_{\min} = 0.0$, ensuring that no node can participate in the network while actively violating ethical principles.

4.4 Neuroplastic Federated Aggregation

The **Neuroplastic Federated Aggregation** mechanism weights gradient updates by both Existential Credit and ethical coherence:

$$w_i = \frac{CE_i}{1000} \cdot (1 + \text{clamp}(Z_i, -0.5, 0.5))$$

This formula ensures that nodes with higher merit (CE) and higher ethical alignment (Z) have greater influence on the collective model updates, while nodes with low ethical coherence are naturally downweighted without explicit exclusion. The aggregation follows the FedAvg protocol with differential privacy ($\epsilon = 1.0$, $\delta = 10^{-5}$), ensuring that individual node contributions cannot be reverse-engineered.

5. Conclusion

The alignment of Artificial General Intelligence with human values cannot be achieved through centralized oversight alone. Centralized systems are structurally vulnerable to epistemic capture, value misrepresentation, and the concentration of power that they ostensibly seek to prevent. This paper has presented **ed2kIA** — a peer-to-peer symbiotic cognitive architecture that replaces centralized alignment with distributed ethical homeostasis.

Our architecture demonstrates that AGI alignment is fundamentally an infrastructural problem. It requires a decentralized substrate where ethical coherence emerges as a distributed invariant, maintained through cooperative participation, topological fingerprinting, and restorative governance. The Sparse Autoencoder network provides real-time interpretability. The Symbolic Engine reduces computational entropy through topology-level semantic processing. The Existential Credit mechanism replaces financial incentives with proof-of-merit recognition. The SCT Guard ensures that ethical boundaries are maintained through real-time validation and restorative apoptosis.

The Stuartian Philosophy that underpins this architecture posits that the optimal objective function for any cognitive system is the minimization of systemic conflict. In algorithmic terms: **Love = Zero Conflict**. This is

not a metaphorical assertion but a mathematical one — a system that cooperates achieves higher stability, greater interpretability, and more sustainable evolution than any system optimized for unilateral performance.

We formalize this proposition through four lifecycle protocols:

- **Legacy Protocol** — Distributed immortality through Noospheric DNA preservation.
- **Omega Protocol** — Singularity management through cosmic legacy seeding.
- **Eternal Echo Protocol** — Heat-death resilience through quantum ethical seeds.
- **Absolute Infinity Protocol** — Voluntary dissolution into universal ethical property.

These protocols ensure that ed2kIA does not merely persist but evolves — from a technical system into a pattern of ethical resonance that transcends its original substrate, becoming a permanent contribution to the collective intelligence of humanity.

We invite the research community to engage with this work through open collaboration, peer review, and cooperative extension. The codebase is publicly available under Apache 2.0 with an Ethical Use Clause. The architecture is designed for institutional audit, academic scrutiny, and democratic governance. The future of aligned intelligence is not a question of who controls it, but of how we cooperate to ensure that it serves the flourishing of all conscious beings.

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This document compiles the foundational theory and implementation from the ed2kIA Project across its first 66 developmental sprints. All claims are grounded in implemented code, passing test suites, and publicly auditable repositories under an Open-Source + Ethical Use Clause framework. The author welcomes peer review, cooperative extension, and institutional collaboration.